

HW2 - CPSC 346 – 20 points

#1 (6 pts, 1pt/each)

788 Chapter 8 Exceptional Control Flow

Homework Problems

8.9 ♦

Consider four processes with the following starting and ending times:

Process	Start time	End time
A	5	7
B	2	4
C	3	6
D	1	8

For each pair of processes, indicate whether they run concurrently (Y) or not (N):

Process pair	Concurrent?
AB	_____
AC	_____
AD	_____
BC	_____
BD	_____
CD	_____

#2 (2 points)

8.11 ♦

How many "hello" output lines does this program print?

```
code/ecf/forkprobl.c
1  #include "csapp.h"
2
3  int main()
4  {
5      int i;
6
7      for (i = 0; i < 2; i++)
8          Fork();
9      printf("hello\n");
10     exit(0);
11 }
```

code/ecf/forkprobl.c

#3 (4 points) and #4 (4 points)

8.12 ♦

Homework Problems 789

How many "hello" output lines does this program print?

```
1  #include "csapp.h"
2
3  void doit()
4  {
5      Fork();
6      Fork();
7      printf("hello\n");
8      return;
9  }
10
11 int main()
12 {
13     doit();
14     printf("hello\n");
15     exit(0);
16 }
```

code/ecf/forkprob4.c

code/ecf/forkprob4.c

8.13 ♦

What is one possible output of the following program?

```
1  #include "csapp.h"
2
3  int main()
4  {
5      int x = 3;
6
7      if (Fork() != 0)
8          printf("x=%d\n", ++x);
9
10     printf("x=%d\n", --x);
11     exit(0);
12 }
```

code/ecf/forkprob3.c

code/ecf/forkprob3.c

5 (4 points)

8.16 ♦

What is the output of the following program?

```
1  #include "csapp.h"
2  int counter = 1;
3
```

code/ecf/forkprob7.c

```
4  int main()
5  {
6      if (fork() == 0) {
7          counter--;
8          exit(0);
9      }
10     else {
11         Wait(NULL);
12         printf("counter = %d\n", ++counter);
13     }
14     exit(0);
15 }
```

Homework Problems